



Evaluation of traditional machine learning algorithms for featuring educational exercises

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Abstract

Artificial intelligence (AI) algorithms are important in educational environments, and the use of machine learning algorithms to evaluate and improve the quality of education. Previous studies have individually analyzed algorithms to estimate item characteristics, such as grade, number of attempts, and time from student interactions. By contrast, this study integrated all three characteristics to discern the relationships between attempts, time, and performance in educational exercises. We analyzed 15 educational assessments using different machine learning algorithms, specifically 12 for regression and eight for classification, with different hyperparameters. This study used real student interaction data from Zenodo.org, encompassing over 150 interactions per exercise, to predict grades and to improve our understanding of student performance. The results show that, in regression, the Bayesian ridge regression and random forest regression algorithms obtained the best results, and for the classification algorithms, Random Forest and Nearest Neighbors stood out. Most exercises in both scenarios involved more than 150 student interactions. Furthermore, the absence of a pattern in the variables contributes to suboptimal outcomes in some exercises. The information provided makes it more efficient to enhance the design of educational exercises.

Keywords Exercise modeling · Machine learning · Classification and regression · Content modeling · Learning analytics

1 Introduction

The growth of AI in education has begun with the development of algorithms for writing and distributing tests and other applications [1]. Such algorithms have been incorporated into models that provide meaningful insights to help teachers perform better in teaching and student learning. The implications of integrating machine learning into the educational sector are vast and varied. This includes new ways to research how machine learning systems can be implemented and develop scenarios to evaluate algorithm efficiency.

One major use of AI in education is Smart Learning Content (SLC) in the sense that the content should be dynamic to fit individual requirements and preferences [21]. Among the different contents, we focused on analyzing educational tests since they are exercises in which students interact with the mentioned types of content. The exercises have other tasks that the student must perform, which can involve multiple attempts, and a mark is given for each attempt. Exercise types encompass multiple choices, multiple responses, drag and drop, and open-ended problems that require manual instructor grading.

The initial hypothesis of this study is that traditional machine learning algorithms can accurately model the relationship between grades, attempts, and time of students when interacting with exercises.

Several machine-learning algorithms have been used to model educational exercises. One of the most common is Item Response Theory (IRT), as indicated by [20]. IRT allows the estimation of item characteristics, such as difficulty, discrimination, and guessing, based on student interactions. Content modeling can also be used to infer the different skills acquired by students through educational materials.

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Key item characteristics recognized through the systematic literature review included the number of attempts, time spent, and grades.

In most studies, such characteristics have helped determine behavioral patterns such as persistence, efficiency, and consistency of intelligent tutoring systems. For instance, [17] adopted indicators, such as the average number of attempts, time spent, and number of completed activities, to analyze student behavior. In education, the speed at which a student acquires certain skills is often referred to as the "learning curve," typically illustrated as 2D graphs that show the relationship between performance (grades) and learning effort (attempts), as well as performance and time spent by [16].

The use of AI algorithms has been presented as a means of enhancing the comprehension of individual and group performance patterns, enabling educators to make well-informed and tactical judgments based on student interactions [14, 15]. This provides a deeper understanding of how these tools can be adapted to different levels of complexity in education by identifying which traditional algorithms perform better based on the characteristics of the exercises. This approach responds to the growing need to personalize each student's learning experience and allows for more targeted and effective support [10, 13].

This study aimed to analyze the possibility of using machine learning methods of regression and classification in educational exercises using real interactions with the students in question. We propose the following research questions:

- RQ1: What traditional machine learning algorithms can be used in an exercise model using the evaluated exercises?
- RQ2: Is it possible to achieve good accuracy using traditional machine learning algorithms in educational exercises?
- RQ3: What characteristics of the dataset most influence the accuracy of each algorithm when characterizing the exercises in a course?
- RQ4: What actions can be taken based on the information obtained from the model exercises?

Answering these four questions is relevant because they allow teachers and practitioners to detect how the exercises are, knowing their features. For example, teachers can detect if an exercise needs to be redone because it is too difficult, if an exercise makes students reflect enough, if the effect of the exercise is different depending on the number of attempts, if the exercise can make students persistent, etc. All of this information would allow teachers to redesign exercises, use exercises in specific contexts, etc. RQ4 is relevant because based on such low-level information, higher teacher actions could be taken. In addition, RQ1, RQ2, and RQ3 are important because models should be as accurate as possible so that

proper decisions can be made, for this reason, it is necessary to know the best machine learning algorithms which might vary depending on the type of exercise (RQ1), the level of accuracy to know the level of certainty of the decisions based on this information (RQ2) and which factors might affect this accuracy (RQ3).

The remainder of this paper is organized as follows. Section 2 summarizes the literature on educational exercise models and their evaluation. Section 3 presents the methodology used in this study. Section 4 presents the results obtained for the two exercises in detail, and, in general, for the rest of the exercises. Section 5 discusses the results and Sect. 6 presents the conclusions, limitations, and future work.

2 Related work

This section presents a review of the literature related to the different exercise models proposed, indicating the inferences made and the educational platforms on which they have been applied. The following subsection presents articles related to the evaluation of exercise models, indicating what type of dataset was used and what metrics were evaluated.

2.1 Exercise model

In the field of machine learning, a model indicates a representation trained on a dataset to perform predictions or analysis [3]. In this context, the exercise model is a combination of exercise features to generate a dataset that is processed using the proposed or traditional machine learning algorithms.

Existing systematic reviews by [48] and [20] describe the different models used in educational exercises. Most studies have focused on predicting the difficulty of the questions or items. However, other exercise characteristics can also be used, such as number of attempts, grade, time spent, discrimination, and guessing.

In this context, Gershon et al. [43] calculated the difficulty of questions in a physics course for different Massive Open Online Courses (MOOCs) using IRT in three different administrations. They found a significant difference in difficulty among the 2015F, 2015S, and 2016S administrations. The distribution of question difficulty in 2015F was lower than in the other two administrations.

Furthermore, Marinho et al. [46] predicted the difficulty parameters of multiple-choice questions in the Brazilian National Secondary Education Exam (ENEM). They used two neural language models and IRT, with the results indicating greater difficulty in differentiating between low-ability students, thus complicating the distinction between their abilities and random choices. The model was also applied to official exam questions to standardize the measurement of

difficulty and to assist in developing questions for a target ability level.

Xue et al. [45] used the rate of incorrect responses to measure the item difficulty. This rate was derived from the proportion of students who responded incorrectly to a particular item, relative to the total number of students interacting with that item. In addition, the authors classified all students into two groups within each domain: those who skipped the domain and those who did not skip the domain. Students who did not skip the domain were further grouped into those who completed the domain and those who responded to some but not all items in the domain.

Using a different approach, Converse et al. [41] estimated item difficulty by comparing traditional IRT methods with the variational autoencoder (VAE) method to estimate item parameters and correlated latent abilities in multidimensional IRT models. The results indicate that the VAE method is computationally efficient for high-dimensional data and produces results comparable to those of traditional parameter estimation methods.

Baral et al. [9] proposed a model called QUICK-Comments to predict open-ended response grades of students in mathematics. This model used sentence-level semantic representations to automate the evaluation of students' open-ended responses. The results showed that the QUICK-Comments model outperformed previous models in terms of accuracy and efficiency when evaluating students' open-ended responses in mathematics.

Finally, Rushkin et al. [44] proposed a parametric statistical model to estimate item response times in online courses. The model was based on a log-normal distribution and was fitted to the user response time data. The results showed that the model could capture variability in response times and provide a good description of the data. In addition, it was demonstrated that the user slowness parameter extracted from the model could predict the success of the user during the course.

2.2 Evaluation of exercise model

To evaluate their different models, researchers used real or simulated students as datasets, depending on the research objective, with different machine learning algorithms and metrics, as in [8]. For example, Xue et al. [45] applied a two-parameter IRT model to estimate the difficulty and discrimination. The authors used question responses from 63625 students from 71 Florida school districts collected during 2017/2018 on the Math Nation platform. The metrics evaluated were the root-mean-square error (RMSE) and absolute bias variance.

Baral et al. [9] proposed a deep learning model designed to predict grades based on student responses to open-ended mathematical questions. This model uses sentence-level semantic representations for analysis. To measure the effectiveness of the model, the researchers compared it to six other Rasch models using machine-learning techniques. They used Pearson's and Spearman's correlation coefficients, mean absolute error (MAE), and mean squared error (MSE) as the evaluation metrics. The dataset included responses from 150,477 students and 2,076 unique questions from 27,199 students. The authors applied the Random Forest and XGBoost algorithms, along with natural language processing techniques, to analyze responses containing both mathematical expressions and nonmathematical text. Furthermore, they conducted a secondary analysis of a dataset of 30,371 student responses collected in 2020 from 12 high school math teachers in spring and fall 2020.

Gershon et al. [43] evaluated the stability of item parameters across multiple administrations of a MOOC using the IRT. Using statistical tests, such as Smirnov's test and ANOVA, the authors determined whether the distributions of the parameters varied significantly among the three courses. The dataset, consisting of 1,278 items and 12,338 students across three semesters, was provided by MITx and included the 'Advanced Introductory Classical Mechanics' course offered in spring 2015 (2015 S), fall 2015 (2015 F), and spring 2016 (2016 S). They applied the IRT model to estimate item difficulty and discrimination parameters for a selected subset of 1,278 items from the initial response matrices.

Hidden for Review [19] used a simulated student context to validate their proposed model and attempted to recreate real-world scenarios. However, there are limitations in creating simulated students that exactly match the behavior of real students. The authors used normal, gamma, and uniform distributions to generate three exercise characteristics: time, grade, and the number of attempts. Furthermore, they created datasets of varying sizes, ranging from 30 to 1,000, to emulate the students' interactions in the exercises.

Dorcca [42] used simulated students to evaluate three strategies within a model of student learning styles, thereby minimizing the resources required to validate the proposed methods, gain insight into the behavior of the proposed system in this context, and make adjustments to improve the design. However, the use of simulated students may not fully represent the behavior and responses of real students, and the effectiveness of adaptive educational systems when tested with simulated students may not always apply to real-world situations.

This study contributes to the evaluation of an exercise model using three characteristics: grade, time, and number of attempts by different artificial intelligence algorithms using real student data, in contrast to the results of our previous study, which utilized student simulations [19].

3 Methods

This section outlines the methodology used to evaluate the exercise model using the quiz dataset. Fig. 1 shows a schematic representation of the experimental procedure followed in the present study, indicating the steps followed. First, we used a real data set provided by Zenodo.org, in which the independent variables were the number of attempts and time spent) and the dependent variable was the grade. Subsequently, we divided the data into a training set (70%) and a test set (30%) and implemented 10-fold cross-validation to evaluate the selected algorithms.

The analyses are carried out using both regression and classification algorithms as they offer complementary perspectives, with different granularity levels. For the case of regression, the objective is to predict the numeric grade of the exercise. For the classification part, the grade is separated into the different categories of qualitative grades that are used in the Spanish System (Excellent, Good, Pass, and Fail), and the objective is to predict the correct grade category. This way, it is possible to analyze how models behave in each case and compare the accuracy of the models when obtaining specific grades and ranges of grades (qualitative grades)

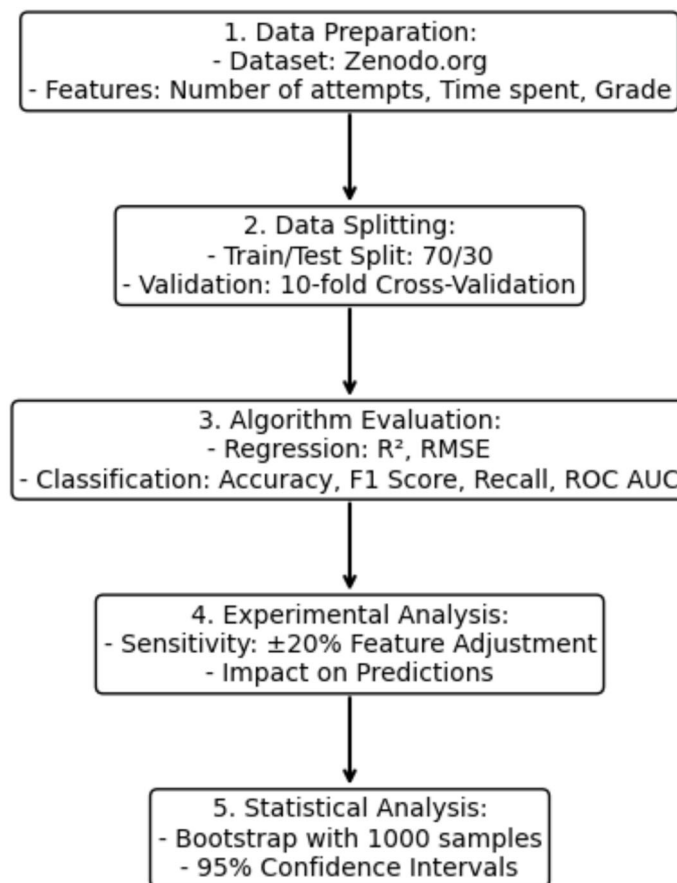
For the regression models, we used R^2 and RMSE as evaluation metrics. For the classification algorithms, we used the accuracy, f1 score, recall, and ROC AUC to evaluate their performance. In addition, we performed a sensitivity analysis, in which we adjusted the independent variables (20% increase and decrease) to evaluate the robustness of the algorithm to small perturbations. Finally, we performed a bootstrap statistical analysis with 1000 samples to generate 95% confidence intervals to measure the uncertainty in the model predictions and ensure that the results were reliable and applicable.

3.1 Design of exercise model

We presented a model design in [47] for educational exercises using features such as the number of attempts and time spent in traditional machine learning algorithms to predict students' grades based on previous interactions. Although we have not used new algorithms, this approach demonstrates how traditional algorithms can be adapted to the needs and effectively applied to analyze and optimize educational exercises using selected features.

Owing to the importance of data-driven decision-making in educational contexts, different predictive models use historical data from previous student interactions [4] providing

Fig. 1 Schematic representation



performance information that allows teachers to make pedagogical decisions to redesign the exercise based on the results obtained in the model. For example, the teacher might detect an exercise that requires a high number of attempts to obtain a good grade or an exercise that has a threshold that no student can pass despite spending more time or attempts [5].

In a previous study [19], we used three feature-generated data simulating previous interactions in the exercise. We then evaluated different machine learning algorithms, such as logistic regression, Nearest Neighbors with different numbers of neighbors, Random Forests with different depths, Decision Trees with different depths, and Gaussian Naive Bayes. The design used time and number of attempts as independent variables and grade as the dependent variable. To represent at least 200 interactions during the exercise accurately, we ran 200 student simulations. Among the different algorithms used, the nearest-neighbor algorithm with k set to 10 showed the highest accuracy. A limitation of this study is the use of simulated students instead of real students, which may affect the generalizability of the results to real educational contexts.

In [47], we used the Jupyter Notebook and Python version 3.9.2, to generate student simulations. We designed an exercise consisting of a multiple-choice question (MCQ) with a correct answer probability of 7%. The simulation involved 300 students, each of whom attempted the exercise at least once. Some students may have attempted the exercise multiple times, resulting in at least 300 interactions. The results of this study revealed certain drawbacks in the use of probabilistic techniques for modeling exercises. The accuracy of the model depends on the amount of data available, which may be low when attempting to model all exercises with a high accuracy. In addition, the model may overlook non-cognitive aspects of learning, such as motivation or learner involvement. Despite these shortcomings, the design of exercise model is a useful tool for improving teaching and learning in different educational contexts and types of exercises. An exercise model was used in this study.

3.2 Dataset

We used the [38] dataset, published in Zenodo.org.¹ This dataset contained different files with information from 158 evaluations of an online undergraduate course. In the files, we can find data about the interactions performed by the students within the evaluations. Among the variables, we have the number of attempts, the time spent on that attempt, and the grade obtained on the same attempt. The information on these three variables can be found in the file `m_quiz_attempts.csv`, which underwent preprocessing and data cleaning before being used as input in the exercise design model to ensure

that the data used in the model training and evaluation were of high quality and in an appropriate format.

This dataset has 158 assessments available, but only 15 were included in this study according to the following criteria to ensure the representativeness of the data. First, only exercises with more than 120 interactions were included, as the number of attempts was an input variable to the algorithm, and a sufficient size of data was guaranteed to perform an efficient analysis, according to a previous study [19]. Second, only exercises with at least four different grades were included to obtain variability in the distribution of the output variable.

3.3 Regression algorithms

In the present study to address the regression problem, we evaluated the model design using a given data set in 12 different machine learning algorithms: Random Forest Regressor [37], Gradient Boosting Regressor [18], Support Vector Regressor (SVR) [24], Lasso Regressor [25], Ridge Regressor [33], ElasticNet Regressor [26], Bayesian Ridge Regressor [23], Huber Regressor [27], K-Nearest Neighbors Regressor [28], Decision Tree Regressor [29], Linear and Logistic Regression [30]. Each of the 12 algorithms was evaluated across five different parameters, resulting in 60 distinct evaluations (12 algorithms $\times$ 5 parameters).

For example, for the Random Forest regressor algorithm in [37], the following parameter combinations were used: the first model with `n_estimators=100` and `max_depth=3` may be a good option for smaller datasets or when looking for a simpler model that avoids overfitting. The second model with `n_estimators=200` and `max_depth=10` provides a greater depth, which allows for more complex modeling and could be beneficial for data sets with more complex patterns. The third model, with `n_estimators=50` and `max_depth=15`, attempts to balance model complexity with a smaller number of estimators. The fourth model with `n_estimators=50` and `max_depth=3` may be useful for simpler data sets, while the fifth model with `n_estimators=150` and `max_depth=5` offers a middle ground in terms of complexity. The different parameters allowed us to evaluate different configurations and determine the optimal balance between the complexity and generalizability of the exercise. Similarly, we combined the different algorithms to determine the best algorithm to fit the model, as shown in Table 16.

3.3.1 Training / testing and cross validation

To evaluate the models, we followed the common practice for machine learning models and split the dataset into training and test sets at a ratio of 70% to 30% [36]. This strategic split allowed for a thorough evaluation of model performance, while ensuring the generalizability of the models to new data.

¹ <https://zenodo.org/records/7300070>

During the training process, 70% of the dataset was used to fit the models, whereas the remaining 30% was reserved to evaluate the performance of each model on unseen data, thereby providing an objective measure of their predictive ability in real-world situations with different training exercises.

To mitigate the risk of overfitting [35] in the 60 combinations of the selected algorithms, we used the `cross_val_score` method of the `sklearn.model_selection` library to evaluate different metrics. A 10-fold cross-validation strategy was implemented by setting `cv_num` to 10, which allowed for a robust evaluation of model performance.

The use of a 10-fold cross-validation has several advantages [34]. First, it provides a more reliable estimate of model performance by averaging the results of ten iterations, thereby reducing the impact of variability in the data. Second, it allowed the detection of potential problems related to data sensitivity and ensured that the model's effectiveness was not dependent on a particular subset. Overall, this approach validated the robustness of the algorithm and increased the confidence in its ability to generalize well to unseen data, strengthening the credibility of our findings in the pursuit of model reliability and effectiveness.

3.3.2 Evaluation metrics

To assess the performance of the models, we used two evaluation metrics from [39]: R-squared (R^2) and Root Mean Squared Error (RMSE) on a scale of 0 to 1. Although these metrics were applied to student models, we used those appropriate for exercise models because in both cases, the models were evaluated to account for variability and precision in the data. R^2 is an essential metric for assessing how well the model explains the variability in the quiz data. RMSE represents the square root of the average squared differences between the predicted and actual values, measuring the typical magnitude of the errors. Together, these metrics offer a comprehensive evaluation of the model's accuracy.

3.3.3 Experimental variation

To evaluate the robustness of the best prediction metric algorithm for sensitivity against alterations in the model input variables, an experimental sensitivity analysis was carried out in which the independent variables were adjusted, the number of attempts and time were increased and decreased by 20%, respectively, to analyze the results related to the prediction of the algorithm. Then, we calculated the average deviation of the predictions concerning an increase or decrease in the features so that a measure of the alteration of the algorithm's results as a function of each variable was achieved. This type of analysis is relevant because it can describe the variables that affect the predictions the most and determine whether the

model is robust to small variations in the input data, which would increase its reliability.

3.3.4 Statistical analysis

To measure the uncertainty in the model's predictions and demonstrate its reliability, we performed a confidence interval analysis using the bootstrap approach. We generated 1000 bootstrap samples by resampling the training data with replacement, trained the model with each sample, and predicted the results on the test set. From these predictions, we calculated the mean prediction and the 2.5th and 97.5th percentiles to establish a 95% confidence interval for each test instance, which provided us with a statistical measure of the variability of the predictions. This ensured that the model results were accurate and reliable when repeatedly sampled. These intervals provide an interpretable range in which the actual prediction can be placed, thereby increasing the confidence in the applicability of the algorithm.

3.4 Classification algorithms

We applied different classification machine learning algorithms to model educational exercises, including K-Nearest Neighbors (KNN), Decision Trees, Random Forests, Logistic Regression, Gaussian Naive Bayes, and Multi-layer Perceptron (MLP). For KNN, we experimented with different k values (1, 2, 3, 5, 10, and 20). By experimenting with different k values, we were able to fine-tune the sensitivity of the model and its ability to generalize based on the training data.

Decision trees with different maximum depths (5, 10, 20, 40, and 60) were used to control the complexity of the model, prevent over-fitting, and improve the generalization ability. We also used random forests with 1000 trees and maximum depths of 3, 5, 10, and 20. Finally, other methods such as logistic regression and Gaussian-naïve Bayes were used.

To capture complex and non-linear relationships, we explored a Multilayer Perceptron (MLP) with ReLU and Tanh activations, which represent artificial neural networks that can capture complex and non-linear relationships in the data. Although other more advanced algorithms, such as Long Short-Term Memory (LSTM) and Convolutional Neural Networks (CNN), may offer additional advantages, our aim in this study is to evaluate a variety of common regression algorithms rather than others with higher complexity and more computational requirements.

3.4.1 Training/testing and cross validations

We employed the same methodology as in the regression analysis. Throughout the training phase, 70% of the dataset was harnessed to fit the models, and the remaining 30%

was reserved for evaluating the performance of each model on unseen data. To mitigate the risk of over-fitting in the classification algorithms, we utilized the `cross_val_score` function from the `sklearn.model_selection` library to evaluate the various performance indicators. We executed a 10-fold cross-validation procedure by setting `cv_num` to 10, which facilitated rigorous evaluation of the model's performance.

3.4.2 Evaluation metrics

To evaluate the performance of the classification algorithms used we have used four metrics [39]: accuracy, f1 score, recall, and roc auc. Precision is the proportion of true positives (TP) among all examples predicted as positive (the sum of true positives and false positives, TP + FP). Recall measures the ability of a model to correctly identify all positive examples. This is the proportion of true positives among all positive examples (the sum of true positives and false negatives, TP + FN/FN).

The F1 Score is the harmonic mean of precision and recall. It provides a single metric that balances both precision and recall and is useful when there is an imbalance between the classes. AUC measures the area under the Receiver Operating Characteristic (ROC) curve, which plots the true positive rate (TPR) versus the positive rate (FPR) at different classification thresholds. A higher AUC value indicates better model performance in terms of its ability to distinguish between the positive and negative classes.

The same experimental variance analysis and statistical analysis were performed for classification.

4 Results

4.1 Regression algorithms

We present a subsection to explain in detail the results of only two specific exercises, while the other subsection provides a general summary of the results of the other exercises.

4.1.1 Detailed analysis of specific exercises

The selected exercise had a range of possible grades from 3.4 to 10.0. Students achieved 20 distinct grade outcomes within this range based on their different attempts. The best resulting algorithm is the Lasso Regressor.

- **Variables:** The exercise was performed by 87 students, with 189 interactions in total. Figure 2 shows the grades of different attempts. Regarding attempts, the students displayed consistent behavior, with an average of 2.1 attempts. Although the minimum number of attempts was one, some students made seven.
- **As for the time spent exercising,** diverse patterns were observed among participants. The average time spent was 18.83 units (units for time were not provided in the dataset), with a minimum of 3.3 units and a maximum of 65.7 units. In terms of performance, the average grade obtained was 8.27, ranging from a minimum of 3.4 to a maximum of 10.
- **Algorithm:** Some performance metrics of the various algorithms used in Exercise 1 are listed in Table 1. The data shows that all algorithms had similar RMSE val-

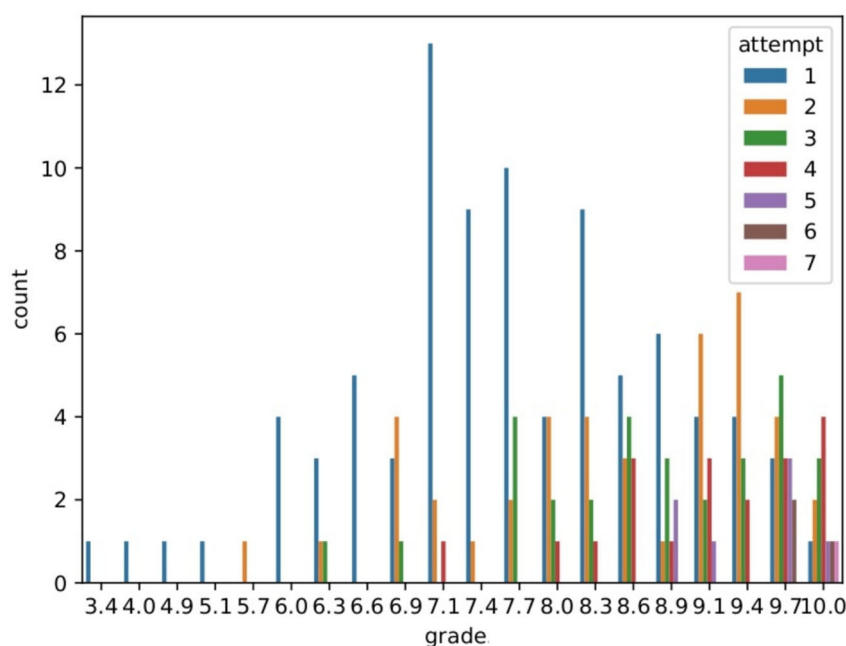


Fig. 2 Variables Exercise 1

Table 1 Metrics of exercise 1

Algorithm	R2	RMSE
Lasso Regressor 4	0.291	0.131
Bayesian Ridge Regressor 1	0.211	0.131
Bayesian Ridge Regressor 2	0.206	0.131
Ridge Regressor1	0.197	0.133

ues, with the focus on R2, which shows that the Lasso Regressor (configured as Lasso Regressor(alpha=0.1, random_state=42)) recorded an R^2 value of 0.291 and a RMSE of 0.131, indicating that 29.1% of the variability in the grades can be accounted for by variations in time and number of attempts.

In summary, although the RMSE values are similar, the R2 metric demonstrates the superior performance of the Lasso Regressor in accounting for the variability in Exercise 1. Therefore, we recommend using the Lasso Regressor for similar tasks, where it is essential to explain the variance of the data. This is particularly important in educational settings, where understanding the influence of variables such as time and the number of attempts on grades can provide useful insights for improving student performance.

Using the best algorithm obtained in this exercise, we performed a more detailed analysis of the predictive ability of the algorithm for different values of the alpha parameter. Table 2 shows the obtained alpha and R2 values. The best result was obtained for an alpha equal to 0.1, with an R2 of 0.291, indicating robust performance and reducing the risk of over-fitting.

No significant R2 improvements were obtained for smaller values of alpha, indicating that the algorithm did not benefit from further regularization. In contrast, we

Table 2 R2 for different alpha values

Alpha	R2
0.00005	0.288
0.0001	0.288
0.0005	0.288
0.005	0.289
0.01	0.289
0.05	0.290
0.075	0.290
0.1	0.291
0.125	0.285
0.2	0.271
0.5	0.108
1	-0.060
5	-0.062

can increase the risk of over-fitting by capturing noise for very low regularization. However, by increasing the value of alpha, a significant decrease was obtained up to negative values with alpha greater than or equal to 1, indicating a decrease in the predictive capability of the model due to over-regularization.

The equations constitute a mathematical system modeling the relationship among the variables "grade," "time," and "attempt" using the Lasso Regressor. Each equation includes coefficients that quantify the relative influence of the variables on the relationship. The specific formulation is as follows:

$$\text{Grade} = 8.235 + (0.295 \times \text{Attempt}) + (0.015 \times \text{Time}) \quad (1)$$

The equations were derived from the algorithm that obtained the best metrics for Exercise 1: the Lasso Regressor. The model was trained using the intercept (lasso_model.intercept_) and coefficients (lasso_model.coef_) derived using this method. These coefficients represent the weights assigned to each variable for predicting grades, time, and attempts in the educational context. (1) describes how a grade is related to the number of attempts and time spent exercising.

- **Experimental Variation:** The evaluation of the sensitivity that has been performed with the Lasso Regressor algorithm, Table 3 shows the results obtained indicating that the variable number of attempts is the most sensitive variable in the algorithm predictions, where the average variation in its sensitivity to a change of $\pm 20\%$ is 0.187, compared to the variable time employed whose average sensitivity was 0.055. This result indicates that the number of attempts is the most influential variable in the model predictions, whereas time has a minor impact.
- **Statistical Analysis:** Table 4 shows the confidence intervals of 95% for the algorithm predictions showing the mean prediction, lower and upper limits of 2.5% and 97.5% respectively, and actual observed values. These results indicate that the prediction assuming a mean prediction is fairly close to reality in most cases but with variability in the predictions captured by the confidence intervals. For example, for an actual value of 6.00, the interval varies between 7.33 and 7.94, which is indicative of some deviation from the model, and values such as 9.40 show tighter intervals between 7.98 and 8.39. However, the Lasso Regressor provided the best performance metrics in this exercise, and it is crucial to

Table 3 Sensitivity analysis

Variable	+20%	-20%
Attempt	0.1874	0.1874
Time	0.0551	0.0551

Table 4 Prediction with confidence intervals

Prediction mean	Lower inter-val (2.5%)	Upper inter-val (97.5%)	Real value
7.64	7.33	7.94	6.00
7.72	7.45	7.97	9.40
8.66	8.50	8.80	9.40
9.18	8.98	9.39	10.00
7.65	7.34	7.94	7.40
8.14	7.87	8.38	8.60
8.32	8.15	8.49	9.70
7.94	7.69	8.19	6.30
8.12	7.83	8.37	8.00
8.20	7.98	8.39	9.40

understand that the appropriateness of a model depends on the specific characteristics of the exercise. Linear models are straightforward and interpretable but may not capture nonlinear relationships effectively; for example, after two attempts, all students obtain the maximum grade in this way, which does not increase the grade linearly with the number of attempts.

The following selected exercise (Exercise 2) obtained the random forest regressor algorithm with the best metric. Therefore, we performed a significant analysis of the characteristics of the independent variables.

- Variables: This exercise was performed by 75 students, with 129 interactions. The average number of interactions per student was 1.72, highlighting the variability in individual participation. Figure 3 shows the distribution

of the grades for different attempts. For the number of attempts, the students exhibited consistent behavior, with an average of two attempts. Although the minimum number of attempts recorded was one, a few students chose to make up to seven attempts, reflecting the different levels of focus and dedication in solving the exercise.

In terms of time spent on the exercise, we observed a variety of paces among the participants. The average time spent was 37.36 time units, with a minimum of 6 units and a maximum of 195 units. These values highlight the diversity of students' time management among students. In terms of performance, the average grade obtained was 8.37, ranging from a minimum of two to a maximum of 10.

- Algorithm: Table 5 lists the performance metrics of Exercise 2 for the different algorithms tested. The average results should reflect the best performance by Random Forest Regressor 2 with configuration `n_estimators = 200`, `max_depth=10`, `random_state = 42`. Thus, the model has the best Root Mean Squared Error among the tested algorithms, with a value of 0.120, which is low and yields higher accuracy in making predictions.

Random Forest Regressor 2 had the highest R-squared value (0.419), which is representative of 41.9% of the variance in grades. This can be explained by the model based on the independent variables, which are time and number of attempts, indicating that almost half of the variability in students' grades was accounted for by the time they spent and the number of attempts.

In contrast, the Gradient Boosting Regressor 5 had an RMSE of 0.128 and an R2 value of 0.291, indicating that while it can predict with reasonable accuracy, it only accounts for 29.1% of the variance in the grades.

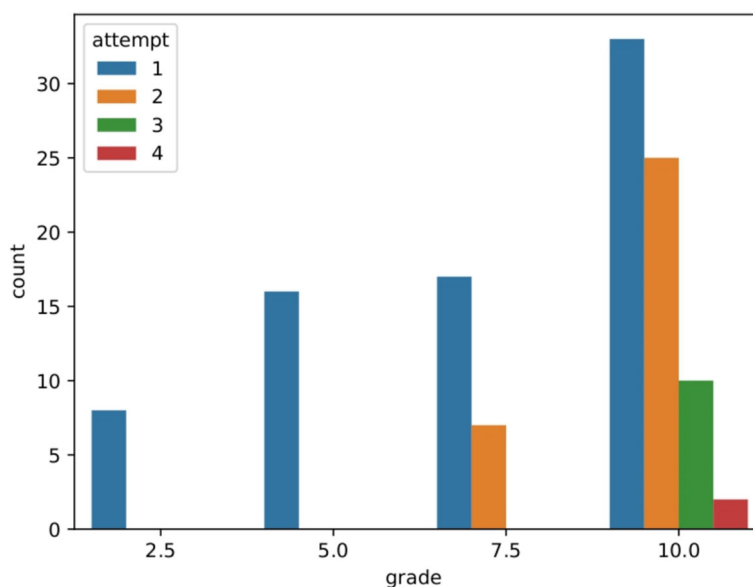
Fig. 3 DataSet Exercise 2

Table 5 Metrics of Exercise 2

Algorithm	R2	RMSE
Random Forest Regressor 2	0.419	0.120
Gradient Boosting Regressor 5	0.291	0.128
Random Forest Regressor 3	0.128	0.135
Random Forest Regressor 5	0.083	0.136
Gradient Boosting Regressor 1	0.202	0.131

This suggests that it is less effective than Random Forest Regressor 2 at capturing the underlying patterns in the data. These findings suggest the importance of considering both the RMSE and R2 metrics when evaluating the model performance. While RMSE provides a measure of predictive accuracy, R2 provides an insight into the explanatory power of the model.

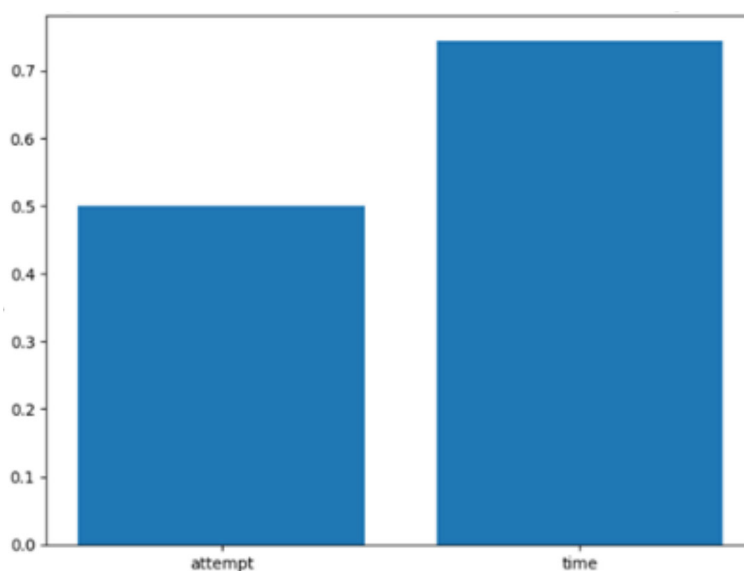
For this exercise, the metrics had high values for the problem of being able to predict the performance of Exercise 2. We would recommend the Random Forest Regressor for similar exercises because it produced excellent results on the metrics evaluated and provided the best explanation of variability for understanding how time and number of attempts affect student grades.

- **Feature importance:** Fig. 4 shows the importance values of the two independent variables, time and number of attempts, obtained using the `feature_importances_` method of the Random Forest Regressor class, which corresponds to the algorithm that obtained the best metrics evaluated in our analysis. According to the results, the importance value assigned to the number of attempts was 0.5, whereas that for the time was 0.73. These values indicate the relative contribution of each variable to the model's predictive ability.

The high importance of the time spent exercising (0.73) highlights its role in the model's predictive capabilities. This result underscores the correlation between the time investment and the grades obtained. However, this relationship raises interesting questions about the dynamics of time management and its direct influence on performance. For instance, while extended time might reflect deeper engagement or effort, it may also signal inefficiency or difficulties in understanding the exercise, suggesting that further analysis of time distribution patterns could uncover latent factors that influence student outcomes.

In contrast, the number of attempts (0.50), while still relevant, contributed less to the model's predictive power. This finding implies that the frequency of interactions is less indicative of prediction than the total time spent. However, the number of attempts cannot be entirely discounted because it can serve as an indicator of persistence or iterative learning processes. A deeper exploration of how attempts interact with other features, such as time spent or specific exercise topology, may provide insights into the nuanced mechanisms driving performance.

The observed metrics of feature importance also raise critical considerations regarding potential biases and assumptions embedded in the grading framework. Although time is a strong predictor, its high importance may inadvertently introduce bias, as longer durations do not always equate to better outcomes. For example, a student who spends more time on an exercise may not necessarily demonstrate greater understanding or performance; rather, they may face challenges that prolong task completion. Similarly, a higher number of attempts may reflect a trial-and-error approach rather than strategic problem-solving.

Fig. 4 Feature Importance Exercise 2

- **Experimental Variation:** The sensitivity evaluation elaborated on the selected algorithm according to Table 6 shows that the time variable is the most sensitive of the predictions. The increase of +20% in the time variable gives an average variation of 0.867, and the decrease of -20% gives 0.636; on the contrary, the number of attempts variable shows a much lower sensitivity effect, with an average variation of 0.0048 for an increase of +20% and 0.0206 for a decrease of -20%. This indicates that time is the most sensitive variable in model predictions, and the number of attempts has a minor impact.
- **Statistical Analysis:** Table 7 shows the 95% confidence intervals of the model, including the mean prediction as well as the lower and upper limits of 2.5% and 97.5%, respectively, in addition to the observed values. The results indicate that the predictions are close to the actual values; however, the confidence intervals also show some variability. Thus, for example, an actual value of 2.5 has a mean prediction of 5.21, with an interval ranging from 2.57 to 8.06, indicating that there is a remarkable range of predictions. In contrast, in the case of an actual value of 10, the confidence intervals were between 9.87 and 10.00, indicating a higher level of determination.

4.1.2 Overview exercises

In the overall analysis of the results obtained for the different exercises, significant variability in performance was observed. Exercise 3 was performed by 80 students who showed consistent performance among themselves, with 129 interactions. The most efficient algorithm for this exercise was Bayes Ridge Regressor 3, which obtained an R2 of 0.015 and an RMSE of 0.193, which was the lowest R2 value compared to the other exercises. This exercise has high variability in the time employed and the number of attempts, which generates noise in the data and decreases the predictive capacity of the algorithm, which is not able to generalize a defined pattern in the exercise.

For Exercise 4, 78 students participated in 139 interactions, achieving an average grade of 8.1, an average number of attempts of 2.12, and an average time of 21.1. Some students solved the exercises quickly, whereas others invested more time in successive attempts to improve their grades. The Random Forest Regressor 4 algorithm was the best predictor of exercise performance, with time as a significant factor. The metrics results were $R^2 = 0.147$ and $RMSE = 0.353$,

Table 6 Sensitivity analysis

Variable	+20%	-20%
Attempt	0.0048	0.0206
Time	0.8671	0.6361

Table 7 Prediction with confidence intervals

Prediction mean	Lower inter-val (2.5%)	Upper inter-val (97.5%)	Real value
8.242	5.404	10.000	10.000
9.964	9.779	10.000	10.000
8.123	5.391	9.972	10.000
6.762	5.780	7.681	7.500
7.045	5.878	8.198	7.500
9.9875	9.8698	10.000	10.000
5.211	2.5719	8.0609	2.500
6.5146	4.9184	7.921	7.500
7.6447	3.061	9.297	10.000
10.000	10.000	10.000	10.000

indicating that the model was useful based on the variability observed.

In addition, in exercises 12, and 14, significant variability in time spent and attempts was also observed. In these exercises, the Random Forest Regressor algorithm proved its efficiency as the best algorithm. For example, in Exercise 12, 107 students performed 176 interactions, obtaining an R2 of 0.251 and an RMSE of 0.22, showing higher prediction accuracy. In exercise 14, with 97 students and 144 interactions, the R2 and RMSE were 0.294 and 0.163, respectively, indicating good model efficiency.

In Exercise 6, with 74 students and 139 interactions, the R2 was 0.027 and the RMSE was 0.559, reflecting low performance due to the dispersion of the variables time and grades. In addition, outliers were found, such as a student who took 49 units to obtain a grade of 0 on the first attempt, but on the second attempt, with a similar time, obtained a grade of 10 after taking 42 units, indicating that time was not directly correlated with the grade. In addition, the number of attempts showed inconsistent patterns, as the grades varied rather than improved. This behavior makes it difficult for the model to identify patterns in the data.

In exercise 5, which had 66 students with 120 interactions, the average grade obtained was 6.8. Despite an average of 1.95 attempts and 62.94 units of time spent per student, initial grades were lower, prompting further effort to improve. SVR 2 was identified as the best algorithm for this exercise, achieving an R2 of 0.016 and RMSE of 0.57. These metrics indicate that although SVR 2 provided predictions, they showed modest accuracy, suggesting that there is room for improvement in predictive ability owing to the considerable RMSE.

Exercise 7 involved an average of 1.52 attempts per student, indicating limited repetition. The best-performing algorithm was the Gradient Boosting Regressor with an R2 of 0.267 and an RMSE of 0.193. The time spent showed significant cases of inversion in exercises 8, 10, 13, and 15

with a mean of 78.41 to 158.67 units. In these cases, the Gradient Boosting regressor was the best algorithm, with R2 values ranging between 0.180 and 0.413, and RMSE values between 0.153 and 0.302, demonstrating its ability to handle data variability and achieve prediction accuracy.

Exercise 9 had an average grade of 8.08, an average of 1,86 attempts, and an average of 22,72 time units per student. This indicates that students obtain a good average grade without spending much time on each attempt. The Lasso Regressor was the best algorithm with an R2 of 0.019 and an RMSE of 0.568, indicating moderate predictive performance. Similarly, Exercise 11 had a mean grade of 8.58, averaging 1,35 attempts per student and 147.70 time units per student. In contrast to the previous exercise, the students had to spend much time getting a good grade on each attempt. The Lasso Regressor algorithm obtained an R2 of 0.103 and an RMSE of 0.513, underlining its ability to provide reliable predictions, despite the difficulties posed by the variability of the data.

Exercise 12 obtained the best performance of the Random Forest model compared to the other exercises, with an R2 of 0.251 and an RMSE of 0.22, due to two factors related to the distribution of the data set and the nature of the algorithm with the indicated parameters. First, the distribution of the grades over the attempts: in the first attempt, the grades were more scattered and lower, while in the following attempts, there was a progressive improvement until the maximum grade was reached. Second, the distribution generates clusters at each attempt, which the Random Forest algorithm detects because of its ability to model nonlinear structures and capture complex patterns. In addition, the parameters $n_estimators=50$ and $max_depth=15$ allow a balance between predictive ability and generalization compared with the other parameter settings evaluated for the algorithm.

Exercise 15 had the best R2 result, with a value equal to 0.435, mainly because of the distribution of grades in each attempt. Most students obtained similar grades on the first attempt, with an average of 8.99, and were grouped. In the second attempt, the number of students dropped to 35 out of 124, but the grades showed uniform improvement, with an average of 9.52. In subsequent attempts, the remaining few students attempted to reach the maximum grade. This concentration in defined groups increases the predictive ability of the algorithm as it reduces the scatter and complexity of the data.

Various algorithms demonstrated optimal performance across the exercises, and the marginal differences in the R2 and RMSE values should be carefully considered. These metrics encompassed R2 values ranging from 0.015 to 0.435 and RMSE values ranging from 0.120 to 0.583, indicating varying levels of predictive accuracy and model fit in different situations. The analysis suggests that although some algorithms show better performance metrics, there is no one-

size-fits-all solution. Variability in algorithm effectiveness highlights the importance of context-specific model selection.

4.2 Classification algorithms

To evaluate the grading algorithms, we converted the grade on a scale from 0 to 10 into a qualitative grade using the Spanish System with the following equivalence: Excellent (EX): grades higher than or equal to 9, Good (GO): grades between 7 and 8.9, Pass (PA): grades between 5 and 6.9, and Fail (FA): grades lower than 5.

4.2.1 Detailed analysis of two exercises

The numbering of the exercises is the same as that described in Table 8, which describes exercise 15 which obtained the best results in the metrics evaluated:

- Variables: Fig. 5 shows the distribution of the qualitative grades EX, GO, PA, and FA as a function of the attempts made. The X-axis shows the four possible attempts, the y-axis represents the number of interactions for each grade, and the bars represent each attempt. There was a clear tendency for the EX grade to predominate in all attempts, especially in the first one, where it exceeded 80 interactions. This trend gradually decreases in subsequent attempts but remains the most frequent grade, even in the fourth attempt, with a single case.

By contrast, the FA grade showed a minimal presence in all attempts, with a maximum of 10 interactions in the first attempt and a single case in the third attempt. The GO and PA grades showed an intermediate behavior, with a higher presence in the first attempt and a gradual decrease in the latter. The behavior of this exercise tends to have a normal distribution with the center at the FA grade, similar to the simulations of the grades used in [19] and [47].

In summary, the results indicated that students tended to get better grades on their first attempt, with a significant prevalence of EX grades. As the attempts progressed, the probability of obtaining this PA grade decreased, whereas the EX grades became dominant again. The FA grade remained low across all attempts, indicating that most students managed to pass the exercise on the first attempt, while a few students who made a fourth attempt achieved a grade.

- Algorithm: Table 9 shows the results of the three best algorithms (Random Forest3, Nearest Neighbors4, and MLP Tanh) with the evaluated metrics and presents a similar performance in classifying qualitative grades, details of the hyperparameters of the algorithms are shown in Table

Table 8 Overview of exercises

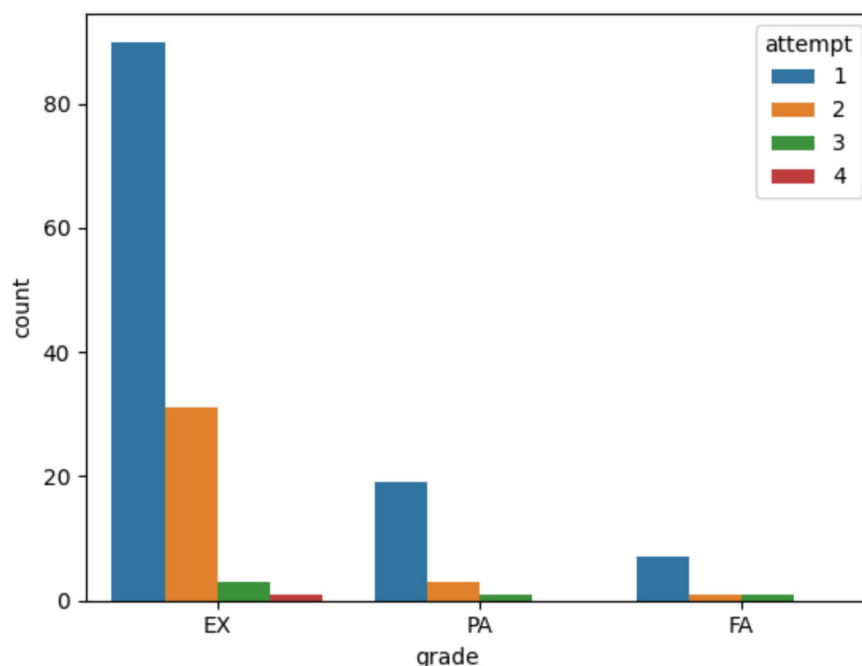
No. of exercise	No. of student	No. of interaction	Avg. grade	Avg. attempt	Avg. time	Best algorithm	R2	RMSE
1	87	189	8,27	2,10	18,83	Lasso Regressor 4	0.291	0.131
2	75	129	8,37	2,03	37,36	Random Forest Regressor 2	0.419	0.120
3	80	129	8,48	1,44	69,77	Bayesian Ridge Regressor 3	0.015	0.583
4	78	139	8,1	2,12	21,1	Random Forest Regressor 4	0.147	0.353
5	66	120	6,8	1,95	62,94	SVR 2	0.016	0.571
6	74	139	2,65	1,67	60,47	Random Forest Regressor 2	0.027	0.559
7	92	134	8,07	1,52	112,54	Gradient Boosting Regressor 3	0.267	0.193
8	83	163	6,66	1,96	78,41	Gradient Boosting Regressor 3	0.043	0.533
9	77	145	8,08	1,86	22,72	Random Forest Regressor 4	0.019	0.568
10	113	133	8,66	1,17	158,67	Gradient Boosting Regressor 4	0.180	0.302
11	87	122	8,58	1,35	147,70	Lasso Regressor 3	0.103	0.513
12	107	176	6,65	1,46	54,46	Random Forest Regressor 3	0.251	0.22
13	94	152	7,19	1,57	95,06	Gradient Boosting Regressor 4	0.413	0.153
14	97	144	8,05	1,38	48,09	Gradient Boosting Regressor 3	0.294	0.163
15	124	156	9,12	1,23	110,79	Random Forest Regressor 4	0.435	0.105

17. Random Forest algorithm had the highest accuracy (0.827) and AUC ROC (0.849), indicating that it is the most effective model for correctly identifying grade and generating a good fit for the receiver operating characteristic (ROC) curve. Nearest Neighbors was the algorithm with the highest F1 score (0.828), suggesting a good balance between accuracy and recall (0.831). NLP Tanh performed similarly to the previous two, with slightly lower precision (0.825) and a similar AUC ROC (0.835). All three algorithms performed well in the classification task, with minimal differences in the metrics evaluated. However, Random Forest was a slightly more favorable

choice because of its higher accuracy and ability to discriminate qualitative grades correctly. Hidden for review [19] recommended using the Nearest Neighbors algorithm for similar exercises based on the results of work using simulated students; the same algorithm is one of the three found in this exercise. Finally, the algorithm's final choice depends on the exercise's specific priorities, such as the importance of the evaluated metrics of the model.

- Experimental Variation: The evaluation of the sensitivity of the best algorithm, according to Table 10, shows that the time variable is considerably more sensitive than the

Fig. 5 Variables Exercise 1



number of attempts in the algorithm's predictions. That is, a 20% increase in the time variable results in an average variation of 0.233, whereas a 20% decrease results in a variation of 0.133. In contrast, no variation was observed with the number of attempts, as the result was identical at 0.033 for both a +20% change and a -20% change. These results indicate that the predictions of the model are conditioned by the time variable, which should be shown to be the most correlated with the result.

- **Statistical Analysis:** Table 11 shows the 95% confidence intervals for the model predictions, including the mean prediction, the lower bound of 2.5%, the upper bound of 97.5% and the actual observations. The confidence intervals of the EX classes were close to the EX, indicating a high correlation and reliability of the predictions. However, in some cases, such as the FA or PA classes, the confidence interval extends to several EX or PA classes, indicating some uncertainty in the predictions. The results also show that the PA class appears at the upper limits of the intervals for several options, indicating that the model has some margin for overestimation or variability in its confidence interval. Although the algorithm can make consistent predictions for the EX class,

it may have difficulty in cases involving less common classes such as PA or FA.

The following describes exercise 14 which obtained the worst results in the metrics evaluated:

- **Variables:** Fig. 6 shows the distribution of qualitative grades segmented by the number of attempts made by students during the exercise. Most students achieved the "PA" grade on the first attempt, with 30 interactions, and the number of students decreased significantly with each subsequent attempt, reaching only two on the fourth attempt.

In contrast, the "EX" grade showed a more even distribution across multiple attempts, with 22 students achieving this grade on the first attempt and 27 on the second attempt. As the number of attempts increased, interactions decreased, but a presence was observed until the fifth attempt, and one case was observed on the seventh attempt. The grade "FA" follows a similar trend to "PA" but with fewer students, where 20 achieved this grade on the first attempt and 16 on the second attempt, with numbers gradually decreasing until the sixth attempt.

In summary, the relationship between the number of attempts and qualitative grades showed no clear trend in

Table 9 Metrics of Exercise 1

Algorithm	Precision	F1 score	Recall	Roc auc
Random forest 3	0.827	0.802	0.781	0.849
Nearest neighbors 4	0.824	0.828	0.831	0.821
Nlp tanh	0.825	0.801	0.781	0.835

Table 10 Sensitivity analysis

Variable	+20%	-20%
Attempt	0.0330	0.0330
Time	0.2333	0.1333

Table 11 Prediction with confidence intervals

Prediction mean	Lower inter-val (2.5%)	Upper inter-val(97.5%)	Real value
EX	EX	EX	EX
EX	EX	PA	EX
EX	EX	EX	EX
EX	EX	PA	PA
PA	EX	PA	FA
EX	EX	PA	EX
EX	EX	EX	EX
EX	EX	EX	EX
EX	EX	PA	EX
EX	EX	EX	EX

specific grades on the first attempt. On the first attempt, students can obtain any of three grades: “PA”, “EX”, or “FA” with relatively high numbers in each category (30, 22, and 20, respectively).

This lack of a clear trend was maintained in subsequent attempts, where multiple grades were possible for the same attempt. For example, on the second attempt, significant numbers are still observed for all three grades: “PA” (9), “EX” (27), and “FA” (16). This pattern of multiple grades associated with any number of attempts suggests considerable variability in student performance, which could complicate accurate and predictable grading of grades.

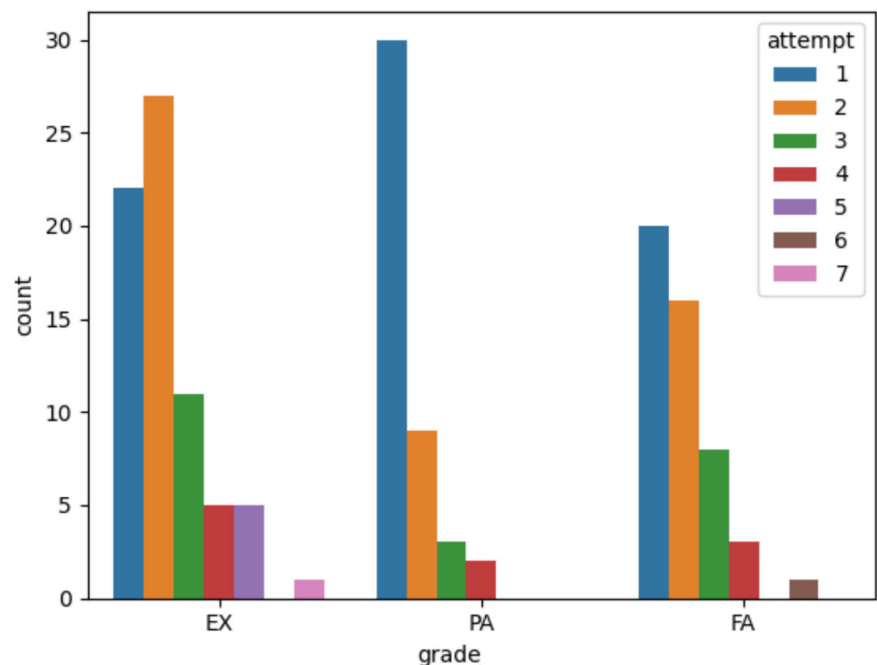
- Algorithm: Table 12 presents the evaluated metrics of the three best algorithms for Exercise 2, showing variations in their effectiveness compared to the previous

Table 12 Metrics of exercise 2

Algorithm	Precision	F1 score	Recall	Roc auc
Random forest 3	0.442	0.439	0.416	0.486
Nearest neighbors 4	0.502	0.513	0.524	0.555
Nlp tanh	0.512	0.508	0.506	0.427

exercise based on the same four metrics previously evaluated. The Random Forest algorithm showed the lowest performance among the three evaluated algorithms, with precision, F1 score, Recall, and ROC AUC values near to 0.5 (0.442, 0.439, 0.416, and 0.486, respectively). In contrast, the nearest neighbor algorithm presented the best performance with an accuracy of 0.502, an F1 score of 0.513, a recall of 0.524, and an ROC AUC of 0.555. Finally, the NLP Tanh algorithm showed an accuracy of 0.512, an F1 score of 0.508, a recall of 0.506, and an ROC AUC of 0.427.

Analysis of the results indicates poor performance in qualitative grade classification due to a relatively uniform distribution between “PA,” “EX” and “FA” grades on the first attempt, making prediction by the algorithm difficult. This lack of a clear pattern persists in subsequent attempts, where the same number of attempts can be associated with different grades, suggesting a significant variability in student performance. These results contrast with those obtained in [19] and [47], which obtained good results in the classification algorithms using normal, gamma, and uniform distributions, but starting from a specific grade, different from the scenario of this exercise.

Fig. 6 Variables Exercise 2

Additionally, no direct correlation was observed between the number of attempts, the grade, and the time spent, indicating that more effort made by the student does not necessarily guarantee a better grade. These findings suggest possible challenges in learning the classification model, which could be attributed to the complexity of the exercise and the lack of clear patterns in student performance. Given these results, there is a need to explore alternative approaches or gather additional information to improve the accuracy of the qualitative grading models.

- **Experimental Variation:** The sensitivity evaluation of the best algorithm, as shown in Table 13, indicates that the time variable has a significantly higher sensitivity than the number of attempts. A 20% increase in the time variable results in an average variation of 0.3125, whereas a 20% decrease leads to a variation of 0.125. For the number of attempts, the sensitivity is much lower, and the variations obtained are 0.051 for a 20% variation and 0.036 for the -20% variation. This indicates that the time variable is predominant in the algorithm's predictions as it has a stronger relationship with performance.
- **Statistical Analysis:** Table 14 shows the 95% confidence intervals for the predictions made by the model, containing the lower bound of 2.5%, the mean prediction, the upper bound of 97.5%, and the actual values. The results indicate that the model obtains relatively consistent predictions for the EX class and that the confidence intervals are in line with the observed values, reflecting an acceptable level of reliability for this class. However, for classes PA and FA, the confidence intervals predicted other classes, which showed greater variability and uncertainty. In certain cases, such as in the upper bound or mean prediction of the PA class, the algorithm tends to overestimate or erroneously confound the actual class, suggesting that the algorithm has problems distinguishing between the PA and EX. The FA class also received inconsistent predictions, whose intervals coincided with other classes, indicating the challenges faced by the model in predicting less frequent classes. These discrepancies indicate a limitation of the algorithm to generalize and the need for improvement, especially in handling unbalanced data.

4.2.2 Overview exercises

Table 15 shows the results of the metrics evaluated in the exercises, highlighting the performance of algorithms such as Random Forest and Nearest Neighbors. In several exer-

Table 13 Sensitivity analysis

Variable	+20%	-20%
Attempt	0.051	0.036
Time	0.3125	0.125

Table 14 Prediction with confidence intervals

Prediction mean	Lower inter-val (2.5%)	Upper inter-val (97.5%)	Real value
EX	PA	PA	PA
EX	EX	FA	EX
EX	EX	PA	EX
EX	EX	EX	EX
EX	EX	PA	FA
EX	FA	FA	EX
EX	EX	EX	FA
EX	EX	PA	EX
EX	EX	EX	EX

cises, Random Forest was generally the algorithm with the best metrics. For example, in Exercise 10, Random Forest obtained an accuracy of 0.809, an F1 score of 0.778, a recall of 0.754, and an impressive AUC of 0.884. Students averaged 1.17 attempts and 158.67 min, with an average grade of 8.66 (GO).

Furthermore, Nearest Neighbor was the best algorithm in the three exercises, and the second best in the remaining exercises, with an average difference of 0.03 in all metrics. Exercise 4, achieved an accuracy of 0.735, an F1 score of 0.727, a recall of 0.72, and an AUC of 0.764, with students making 1.73 attempts and spending 21.1 min per exercise, resulting in an average grade of 8.1 (GO). However, in Exercise 8, Nearest Neighbors performed less effectively, with an accuracy of 0.532, F1 score of 0.528, recall of 0.526,

Table 15 Overview exercises

Number of exercise	Best algorithm	Precision	F1 score	Recall	Roc auc
1	Random Forest 3	0.588	0.56	0.534	0.611
2	Random Forest 3	0.721	0.694	0.669	0.706
3	Random Forest 4	0.792	0.784	0.766	0.776
4	Nearest neighbors 4	0.735	0.727	0.72	0.764
5	Nearest neighbors 4	0.618	0.621	0.624	0.631
6	Random Forest 3	0.787	0.75	0.717	0.731
7	Random Forest 3	0.753	0.715	0.682	0.746
8	Nearest neighbors 4	0.532	0.528	0.526	0.522
9	Random Forest 4	0.713	0.664	0.646	0.712
10	Random Forest 3	0.809	0.778	0.754	0.884
11	Random Forest 3	0.781	0.749	0.72	0.752
12	Random Forest 3	0.545	0.574	0.608	0.613
13	Random Forest 3	0.562	0.565	0.568	0.588
14	Random Forest 3	0.442	0.439	0.416	0.486
15	Random Forest 3	0.827	0.802	0.781	0.849

and AUC of 0.522. Students averaged 1.96 attempts and 78.41 min, earning an average grade of 6.99 (PA).

In some exercises, the performances of both Random Forest and Nearest Neighbors were lower than the metrics evaluated. For example, in exercise 1, Random Forest obtained an accuracy of 0.588, an F1 score of 0.56, a recall of 0.534, and an AUC of 0.611. Similarly, in exercise 12, Random Forest obtained an accuracy of 0.545, an F1 score of 0.574, a recall of 0.608, and an AUC of 0.613. In exercise 13, Random Forest obtained an accuracy of 0.562, an F1 score of 0.565, a recall of 0.568, and an AUC of 0.588. In these exercises, there is no defined behavior for the three variables of the model; therefore, the results obtained are unsuitable for grade classification.

In summary, the results show that after evaluating the 28 different classification algorithms with their respective parameters, the Random Forest3 algorithm provided the best values in the evaluated metrics, where normal behavior was observed in the qualitative grade variable with a predominant class, showing robustness and consistency in the ability to distinguish between classes. Moreover, the Nearest Neighbors4 algorithm performed well in the same exercises, but with an insignificant difference of 0.03 concerning the Random Forest3 metrics. In a few exercises, it obtained better results in the metrics. Finally, in exercises where there was no predominant class behavior, it was not possible to obtain good results for all evaluated algorithms.

5 Discussion

In this section, we examine the results of the exercises to answer the research questions.

5.1 RQ1: What traditional machine learning algorithms can be used in an exercise model using the evaluated exercises?

The results showed that the regression and classification algorithms performed differently depending on the exercise characteristics, focusing on the number of attempts, time spent, and grade obtained.

In exercises where students showed a tendency to make multiple attempts to achieve a high grade, the Random Forest Regressor and Gradient Boosting Regressor algorithms stood out for their ability to capture the complexity of the data and accurately predict final grades. This suggests that these algorithms are particularly effective in exercises that require an iterative learning process on the part of the students. In addition, these algorithms have been used in other predictive models using e-learning data, such as in predicting school dropouts [31] and predicting students' academic performance [32].

In exercises where students demonstrated greater efficiency by achieving results with fewer attempts and less time spent, classification algorithms such as Random Forest and Nearest Neighbors performed well in predicting qualitative grades. The results also showed significant variations in the performance of the algorithms depending on the distribution of the grades and the complexity of the data.

In summary, the selection of a machine learning algorithm should be tailored to the specific characteristics and predictive goals of each educational exercise. Regression algorithms, such as Random Forest and Gradient Boosting, are suitable for handling data complexity in exercises with multiple attempts and variability in time spent. In contrast, classification algorithms such as Random Forest and Nearest Neighbor are effective for exercises with clearer and more predictable distributions. In [20], the results indicated that these algorithms are the most used among the different content models proposed.

5.2 RQ2: Is it possible to achieve good accuracy using traditional machine learning algorithms in educational exercises?

Different metrics have been used to evaluate the design of an exercise model, such as the accuracy, precision, F1-score, AUC, RMSE, and R2, as reported by [39].

We used metrics, such as R-squared and RMSE, to evaluate the performance of the regression models. R2, as in [22], uses the R2 metric to estimate the total variance of each predictor variable using artificial neural networks. Similarly, RMSE was used by [41] and three different parameter estimation techniques and datasets were evaluated to determine the smallest possible error using the RMSE metric. In addition, [40] evaluated the model using the RMSE metric calculated between expected and observed mean grades.

Among the algorithms evaluated, the Lasso Regressor and the Bayesian Ridge Regressor proved to be the most effective in predicting student grades as a function of variables such as attempts and time spent on the exercise. The complexity of the exercise was found to influence model performance, as linear models performed well with clear relationships between variables but did not effectively capture nonlinear relationships. The time spent exercising emerged as a significant factor in model predictions, indicating its importance in student performance, whereas the number of attempts was less important.

However, the performance of the classification algorithms was exercise-dependent, and the random forest and nearest neighbor algorithms demonstrated good overall performance. For the classification exercises, the precision ranged from 0.442 to 0.827, the F1 score ranged from 0.439 to 0.802, the recall ranged from 0.416 to 0.781, and the AUC ROC ranged from 0.486 to 0.849. These results indi-

cate that, although the predictive power of the algorithms can vary considerably, it is possible to obtain high accuracy and good performance in some exercises. Hidden for review [19] recommended using the nearest neighbor algorithm in similar educational exercises, obtaining precisions of 0.758 and 0.878, although he used similar, not real, students.

In the regression analysis, the R^2 values ranged from 0.015 to 0.435, whereas the RMSE values ranged from 0.120 to 0.583. These results indicate that, although the algorithms may have a limited ability to explain variability in some cases (low R^2), it is possible to achieve relatively low prediction errors (low RMSE) in others. These findings suggest that the effectiveness of both the classification and regression machine learning algorithms depends significantly on the design of the exercise and the specific characteristics of the data being evaluated.

While it is difficult to determine the specific reasons for the behavior of each model, the fact that the Random Forest behaves well in most of the contexts is consistent with previous works in the literature where Random Forest outperformed other methods [49, 51], and it is one of the most common algorithms in the literature [50]. Nevertheless, further work could be done to better understand the reasons behind the performance of each algorithm.

In addition, to evaluate the reliability of the algorithms, statistical and experimental variance analyses were performed, including a sensitivity analysis and confidence intervals for the predictions. The sensitivity analysis showed that certain variables, such as the time spent performing the exercises, had a greater influence on the predictions than the number of attempts, suggesting that time and performance test results were more correlated.

Furthermore, the confidence intervals of the predictions showed that the mean data of the predictions were generally close to the observed values for the most frequent classes and that these intervals widened for less common classes, such as the PA and FA classes, suggesting greater uncertainty in these cases.

5.3 RQ3: What characteristics of the dataset most influence the accuracy of each algorithm when characterizing the exercises in a course?

In [19], we recommend using at least 200 interactions in an exercise to obtain an average accuracy of 0.8. However, in the exercises evaluated, the average number of interactions was 150, with a minimum of 129 and a maximum of 206. Interestingly, the exercise with the highest number of interactions (206) had a relatively low accuracy of 0.588, whereas the exercise with the lowest number of interactions (129) had a high accuracy of 0.792. These results suggest that, for these specific exercises, there is no direct relationship between the number of interactions and the accuracy of the algorithm.

We found a relationship between the distribution of grades, time, and the number of attempts. In exercises in which the distribution of grades was more uniform over multiple attempts, the accuracy of the algorithm was greater than 0.7. In contrast, in exercises where performance was below 0.6, the distribution of grades was skewed toward certain attempts, indicating that there was no clear pattern between the dependent variables and the grade obtained. This suggests that a uniform distribution of grades across attempts may contribute to greater accuracy in machine learning models.

In conclusion, although the number of interactions does not appear to be a determining factor for the accuracy of the algorithm, the distribution of grades across multiple attempts was. A uniform distribution of grades improves the ability of the algorithm to make accurate predictions, highlighting the importance of considering the distribution of grades across interactions and times to improve model performance.

5.4 RQ4: What actions can be taken based on the information obtained from the model exercises?

Analysis of the results highlights several strategic actions that can be taken to improve the effectiveness of the exercises. Therefore, time management is crucial. Wolters and Brady [6] found that students who manage their time well tend to achieve better academic performance.

In Exercise 4, the pattern of time management was shown to be an important performance factor. For example, it was observed that some students obtained high grades in short times such as 6.4 units, while other students required more time 30 min to obtain the same performance. In addition, different time patterns were identified based on the number of attempts. In the first attempt, students show more variable times from 4.4 to 72.5, while in subsequent attempts, a reduction of time is observed in ranges between 6.4 min and 15 min, and most of them obtained a high grade.

These findings highlight the importance of providing specific guidance on time management and encouraging iterative learning practices to optimize both academic performance and task-solving efficiency.

Second, Exercise 6 has a positive learning curve, but with an initial difficulty, many students (70%) obtained a grade of 0 on the first attempt, regardless of the time invested, indicating that the exercise is challenging owing to the high complexity of the problem or lack of adequate strategies.

However, this difficulty was overcome on the second attempt, when most students (60%) obtained the maximum grade, indicating that students persevered but could benefit from more detailed feedback and incorporating specific resources, such as additional practice exercises or tutorials, to provide them with the extra help they needed to overcome challenges and improve their [7].

Exercise 3 was a very easy pattern, so most students got the highest grade (10 points), with an average grade of 1.23 points, most on the first attempt, and an average time of 79 units. The exercise is too simple, which may limit students' ability to develop more advanced skills, such as problem-solving or critical thinking. In addition, the low number of attempts reflects a lack of motivation to explore further, suggesting that the exercises should be well designed to challenge students without being excessively difficult [2].

In summary, to optimize students' learning experiences, it is essential to provide specific guidance on time management, persistence, and perseverance; adapt course orchestration to different learning styles; provide additional resources when needed; and design exercises that encourage persistent effort.

6 Conclusion and future work

The findings from this study provide valuable insights for enhancing the effectiveness of exercises in various situations, considering three characteristics: number of attempts, grade obtained, and time spent. First, the clear association between the cognitive demands of the exercises and the performance of traditional machine learning algorithms highlights the importance of adapting the choice of algorithm to the complexity of the exercises. For regression, we found that the Bayesian Ridge Regressor performed well on tasks where students could obtain a good grade quickly. For tasks that required students to invest time to obtain a good grade, the Random Forest regressor performed well. In contrast, for tasks that required multiple attempts, the Gradient Boosting and Random Forest Regressor algorithms performed the best for the metrics evaluated.

Furthermore, traditional classification algorithms such as Random Forest and Nearest Neighbors [19] obtained good results for the evaluated metrics. However, performance was not uniform across all exercises. A significant decrease in accuracy was observed in some cases, mainly owing to class imbalance. Based on these findings, a thorough analysis of data and evaluation metrics is recommended before selecting a specific classification algorithm. In addition, future research should explore preprocessing techniques to optimize the performance and robustness of these algorithms.

The results of this study can be used to improve educational systems. Some examples are the following: 1) exercises for which the maximum grade is difficult to achieve can be detected, which can imply a redesign of the materials; 2) corrective messages can be included for exercises that do not reflect at the beginning; 3) design of hints can be introduced for exercises for which it is difficult to make progress at the beginning even if students devote time; and 4) personalized strategies can be designed for exercises for which different clusters of student behavior are detected.

The following strategies are recommended to improve the effectiveness of educational exercises based on the analyzed characteristics. In exercises that require students to invest a lot of time to achieve a good grade, a redesign focused on exercise topology could improve grades, allowing for more interaction and learning in line with [6]' recommendations.

Exercises that require multiple attempts to solve are not inherently problematic but can serve as indicators for educators to review and potentially redesign these tests. For example, a combination of redesign, adjustment of the three proposed features (time, number of attempts, and grade), and the provision of additional support could make them more effective and efficient in terms of the time spent by students. This approach is supported by research that highlights the importance of personalized support and repeated practice to improve learning. This approach is supported by research highlighting the importance of personalized support and repeated practice in the learning of [7].

A limitation of the present study was the use of a specific dataset for the evaluation of the algorithms, which did not allow us to generalize the results to all educational exercises using the three selected features. However, the results obtained allowed us to illustrate the performance of the model and propose specific algorithms for different scenarios that could be used by other researchers to evaluate the same characteristics using other datasets. Furthermore, the reduced number of exercises contributes to the previous limitation, although one of the conclusions is that the exercise design has a strong impact on the model. Thus, if wide differences are found between these exercises, it is expected that they will also appear with new exercises. In relation to this, it would be interesting to classify exercises into clusters depending on the design and determine if the results are more homogeneous within each cluster.

In future work, to reduce the complexity of the dataset, we will use the Kolmogorov method and analyze whether it significantly improves the predictive ability of the algorithms using the same educational exercises, although some were not as efficient owing to the distribution of the data. Previous studies [11, 12] used this approach to make models identify relevant patterns more efficiently, reduce the possibility of overfitting, and improve generalization.

We intend to evaluate the model in another real scenario with a larger number of students performing educational exercises to determine its generalizability and to include other characteristics of the exercises, such as the number of answer options, type of question, and level of difficulty, among others, to analyze whether there is an improvement in the performance of the algorithms and whether this improvement is not correlated with the three initially proposed [47]. Moreover, in terms of the algorithms, more work is needed to better explain why each model behaves better or worse in each context and the specific characteristics that affect

the algorithms. Furthermore, more algorithms can be tested, particularly more complex deep learning models with neural networks.

Using these three features, we will create dashboards of explainable visualizations using the information obtained from the model and the datasets. Visualizations should allow the instructor to identify patterns in the exercises or possi-

ble problems and make the necessary corrections, including redesigning the exercise or orchestrating the course, to improve the teaching-learning process for students.

7 Appendix A List of algorithms used with their hyperparameters

Table 16 Regression models

Name of algorithm	Algorithm	Hyperparameters
Random Forest Regressor 1	RandomForestRegressor	n_estimators=100, max_depth=3 random_state=42
Random Forest Regressor 2	RandomForestRegressor	n_estimators=200, max_depth=10 random_state=42
Random Forest Regressor 3	RandomForestRegressor	n_estimators=50, max_depth=15 random_state=42
Random Forest Regressor 4	RandomForestRegressor	n_estimators=50, max_depth=3 random_state=42
Random Forest Regressor 5	RandomForestRegressor	n_estimators=150, max_depth=5 random_state=42
Gradient Boosting Regressor 1	GradientBoostingRegressor	n_estimators=100, learning_rate=0.1 max_depth=3, random_state=42
Gradient Boosting Regressor 2	GradientBoostingRegressor	n_estimators=200, learning_rate=0.1 max_depth=5, random_state=42
Gradient Boosting Regressor 3	GradientBoostingRegressor	n_estimators=50, learning_rate=0.01 max_depth=3, random_state=42
Gradient Boosting Regressor 4	GradientBoostingRegressor	n_estimators=100, learning_rate=0.01 max_depth=5, random_state=42
Gradient Boosting Regressor 5	GradientBoostingRegressor	n_estimators=150, learning_rate=0.05 max_depth=4, random_state=42
SVR 1	SVR	kernel='rbf', C=1
SVR 2	SVR	kernel='linear', C=0.1
SVR 3	SVR	kernel='poly', degree=2, C=1
SVR 4	SVR	kernel='rbf', C=10, gamma=0.1
SVR 5	SVR	kernel='sigmoid', C=0.01, coef0=0.1
Lasso Regressor 1	Lasso	alpha=0.01, random_state=42
Lasso Regressor 2	Lasso	alpha=0.001, random_state=42
Lasso Regressor 3	Lasso	alpha=1.0, random_state=42
Lasso Regressor 4	Lasso	alpha=0.1, random_state=42
Lasso Regressor 5	Lasso	alpha=0.0001, random_state=42
Ridge Regressor 1	Ridge	alpha=1.0, random_state=42
Ridge Regressor 2	Ridge	alpha=0.1, random_state=42
Ridge Regressor 3	Ridge	alpha=0.01, random_state=42
Ridge Regressor 4	Ridge	alpha=0.001, random_state=42
Ridge Regressor 5	Ridge	alpha=10.0, random_state=42
ElasticNet Regressor 1	ElasticNet	alpha=0.01, l1_ratio=0.5 random_state=42
ElasticNet Regressor 2	ElasticNet	alpha=0.1, l1_ratio=0.3 random_state=42

Table 16 continued

Name of algorithm	Algorithm	Hyperparameters
ElasticNet Regressor 3	ElasticNet	alpha=0.001, l1_ratio=0.7 random_state=42
ElasticNet Regressor 4	ElasticNet	alpha=1.0, l1_ratio=0.2 random_state=42
ElasticNet Regressor 5	ElasticNet	alpha=0.0001, l1_ratio=0.8 random_state=42
Bayesian Ridge Regressor 1	BayesianRidge	compute_score=True
Bayesian Ridge Regressor 2	BayesianRidge	alpha_1=1e-6, alpha_2=1e-6 compute_score=True
Bayesian Ridge Regressor 3	BayesianRidge	alpha_1=1.0, alpha_2=1.0 compute_score=True
Bayesian Ridge Regressor 4	BayesianRidge	lambda_1=1e-6, lambda_2=1e-6 compute_score=True
Bayesian Ridge Regressor 5	BayesianRidge	lambda_1=1.0, lambda_2=1.0 compute_score=True
Huber Regressor 1	HuberRegressor	Nothing
Huber Regressor 2	HuberRegressor	epsilon=1.0
Huber Regressor 3	HuberRegressor	epsilon=1.5
Huber Regressor 4	HuberRegressor	max_iter=100
Huber Regressor 5	HuberRegressor	alpha=0.01
K-Nearest Neighbors Regressor 1	KNeighborsRegressor	n_neighbors=5
K-Nearest Neighbors Regressor 2	KNeighborsRegressor	n_neighbors=7, weights='distance'
K-Nearest Neighbors Regressor 3	KNeighborsRegressor	n_neighbors=3, weights='uniform'
K-Nearest Neighbors Regressor 4	KNeighborsRegressor	n_neighbors=10, weights='uniform'
K-Nearest Neighbors Regressor 5	KNeighborsRegressor	n_neighbors=5, weights='distance'
Decision Tree Regressor 1	DecisionTreeRegressor	max_depth=5, random_state=42
Decision Tree Regressor 2	DecisionTreeRegressor	max_depth=10, random_state=42
Decision Tree Regressor 3	DecisionTreeRegressor	max_depth=15, random_state=42
Decision Tree Regressor 4	DecisionTreeRegressor	min_samples_split=5, random_state=42
Decision Tree Regressor 5	DecisionTreeRegressor	min_samples_leaf=3, random_state=42
Linear Regression 1	LinearRegression	Nothing
Linear Regression 2	LinearRegression	fit_intercept=True, normalize=True
Linear Regression 3	LinearRegression	fit_intercept=False
Linear Regression 4	LinearRegression	fit_intercept=False, copy_X=True
Linear Regression 5	LinearRegression	fit_intercept=True, copy_X=False
Logistic Regression 1	LogisticRegression	Nothing
Logistic Regression 2	LogisticRegression	C=0.1
Logistic Regression 3	LogisticRegression	solver='liblinear'
Logistic Regression 4	LogisticRegression	max_iter=1000
Logistic Regression 5	LogisticRegression	penalty='l1', solver='liblinear'

Table 17 Classification Models

Name of algorithm	Algorithm	Hyperparameters
Nearest Neighbors 1	KNeighborsClassifier	n_neighbors=2
Nearest Neighbors 2	KNeighborsClassifier	n_neighbors=3
Nearest Neighbors 3	KNeighborsClassifier	n_neighbors=5
Nearest Neighbors 4	KNeighborsClassifier	n_neighbors=10
Nearest Neighbors 5	KNeighborsClassifier	n_neighbors=20
Decision Tree 1	DecisionTreeClassifier	max_depth=5
Decision Tree 2	DecisionTreeClassifier	max_depth=10
Decision Tree 3	DecisionTreeClassifier	max_depth=20
Decision Tree 4	DecisionTreeClassifier	max_depth=40
Decision Tree 5	DecisionTreeClassifier	max_depth=60
Random Forest 1	RandomForestClassifier	n_estimators=1000, max_depth=3
Random Forest 2	RandomForestClassifier	n_estimators=1000, max_depth=5
Random Forest 3	RandomForestClassifier	n_estimators=1000, max_depth=10
Random Forest 4	RandomForestClassifier	n_estimators=1000, max_depth=20
Random Forest 5	RandomForestClassifier	n_estimators=1000, max_depth=40
Gradient Boosting 1	GradientBoostingClassifier	n_estimators=100, learning_rate=0.1 max_depth=3, random_state=42
Gradient Boosting 2	GradientBoostingClassifier	n_estimators=200, learning_rate=0.1 max_depth=5, random_state=42
Gradient Boosting 3	GradientBoostingClassifier	n_estimators=50, learning_rate=0.01 max_depth=3, random_state=42
Gradient Boosting 4	GradientBoostingClassifier	n_estimators=100, learning_rate=0.01 max_depth=5, random_state=42
Gradient Boosting 5	GradientBoostingClassifier	n_estimators=150, learning_rate=0.05
SVR 1	SVR	kernel='rbf', C=1
SVR 2	SVR	kernel='linear', C=0.1
SVR 3	SVR	kernel='poly', degree=2, C=1
SVR 4	SVR	kernel='rbf', C=10, gamma=0.1
SVR 5	SVR	kernel='sigmoid', C=0.01, coef0=0.1
Gaussian Naive Bayes	GaussianNB	Nothing
MLP (ReLU)	MLPClassifier	hidden_layer_sizes=(100, 100, 100) max_iter=10000, activation='relu'
MLP (tanh)	MLPClassifier	hidden_layer_sizes=(100, 100, 100) max_iter=10000, activation='tanh'

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